

# RIFAT ERDEM SAHIN

## SENIOR DEVOPS ENGINEER — NGINX, RHEL PLATFORM SECURITY & CNI MIGRATION

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 **Location:** London, United Kingdom  **Citizenship:** British  **Email:** [contact@rifaterdemsahin.com](mailto:contact@rifaterdemsahin.com)   
**Phone:** +44 7848 024173  **LinkedIn:** [linkedin.com/in/rifaterdemsahin](https://www.linkedin.com/in/rifaterdemsahin)  **GitHub:**  
[github.com/rifaterdemsahin](https://github.com/rifaterdemsahin)  **Portfolio:** <https://rifaterdemsahin.com>  **Schedule a Call:**  
<https://calendly.com/rifaterdem/schedule>

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## ELIGIBILITY & CLEARANCES

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**Work Authorization:** Eligible to work in the UK — no sponsorship required **Contract Types:** Inside IR35, Outside IR35, PAYE **Security Clearance (Active):** NATO · SC (Security Check) · CNI (Critical National Infrastructure) · BPSS · NSV **Background Check Sponsors:** Sterling, IBM, Capgemini **Availability:** Immediate — currently open to work

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## PROFESSIONAL SUMMARY

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Senior DevOps Engineer with 12+ years delivering and hardening Linux platforms for Critical National Infrastructure (CNI), finance, and enterprise clients — most recently on RHEL/OpenShift estates at **National Grid** and **NESO (National Energy System Operator)**. Direct, hands-on delivery experience across every technology named in this role: **NGINX** as a production reverse proxy/load balancer, **F5** perimeter WAF policy operation (AppProtect-class rule sets), **OPSWAT MetaDefender** AV/AM scanning integration, **SELinux**-enforced RHEL/CentOS hardening, **Ansible** configuration automation, daily **Bash** and **Python** scripting, and 10+ years of **Azure DevOps**/CI-CD pipeline delivery. Comfortable owning a NGINX/RHEL migration programme end-to-end — from restart and re-scoping through secure, auditable delivery in a live CNI environment. Active **NATO**, **SC**, and **CNI** clearances.

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## TARGETED SKILLS — WHERE I'VE USED THEM

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This section maps each requested skill directly to the engagement where I used it, not just a keyword list.

### **NGINX**

At **Red Box Recorders**, I architected and deployed **NGINX** as the production reverse proxy and load balancer in front of a CQRS microservices platform (EventStore, Redis, Minio) running on an on-premise Kubernetes cluster (VMware/CentOS). NGINX handled SSL termination, path-based routing, and traffic cutover during the migration from Docker Compose to Kubernetes, operating alongside MetalLB and

Traefik for ingress. I've continued to use NGINX as a lightweight edge proxy in later contract and personal-infrastructure work for routing and TLS termination in front of containerised services.

## **F5 (AppProtect-class Web Application Firewall)**

At **National Grid** and **NESO**, F5 appliances sat in the daily-operated perimeter security stack protecting CNI-hosted OpenShift/Kubernetes platforms, alongside Zscaler and Checkpoint. I operated and maintained WAF policy sets functionally equivalent to F5 AppProtect — inspecting and blocking malicious HTTP traffic before it reached CNI production services — and coordinated every policy change through formal change control (eChips at NESO) to avoid disrupting national energy-balancing systems. This is daily, hands-on WAF operations in a regulated CNI context, not a one-off proof of concept.

## **OPSWAT MetaDefender (AV/AM)**

At **NESO / National Grid**, I integrated **OPSWAT MetaDefender** into the CNI ingress and file-transfer pipeline alongside F5 and Zscaler to provide multi-engine antivirus/anti-malware scanning of files and payloads moving from lower Azure test environments into the on-premises CNI production estate. This scan-before-promote control was a mandated part of the CNI security baseline — content wasn't allowed to cross the environment boundary without a clean MetaDefender result.

## **SELinux**

At **Red Box Recorders**, I hardened the on-premise CentOS hosts underpinning the Kubernetes cluster with **SELinux running in enforcing mode**, rather than the common shortcut of disabling it. I defined custom policy contexts/booleans so the CQRS services could run under mandatory access control without breaking functionality, reducing the container-breakout attack surface on an on-prem estate that had no cloud-provider isolation to fall back on.

## **Ansible**

At **National Grid**, I used **Ansible playbooks** to automate RHEL server build, patching, and configuration-drift remediation across the OpenShift/VMware estate, replacing manual runbook execution and cutting environment provisioning time. Ansible sat alongside Tekton and Argo-CD in the GitOps toolchain, handling the host-level configuration layer that Kubernetes manifests don't cover.

## **Bash**

Bash has been my day-to-day Linux operations language throughout my career. At **Red Box Recorders**, I wrote Bash scripts to drive the Kubernetes/Docker-Compose migration cutover — service start/stop ordering, health-check polling, and log capture. Across CNI and finance engagements since, I've maintained Bash tooling for SSH-based remote administration, log triage, and CI pipeline glue scripts.

## **Python**

I've used **Python** for DevOps automation and debugging across multiple engagements: at **Concentra Analytics**, for AWS Lambda-based serverless automation and pipeline tooling; at **Smart Space Software**, for scripting and configuration validation supporting a Service Fabric migration; and more recently, for infrastructure automation scripts and log/telemetry processing feeding CI/CD pipelines.

## Azure DevOps / DevOps (general)

Azure DevOps (VSTS) has been the constant thread across my contracting career — 10+ engagements since 2016, including **NESO** (Azure DevOps Pipelines orchestrating GitOps delivery to OpenShift), **Goldman Sachs**, **Ypsomed**, **Cushman & Wakefield**, **Emerson**, **Canada Life**, **Theta Trading**, **Sword**, **Craneware**, **Scottish Enterprise**, and **Microsoft**. I design and maintain both classic and YAML pipelines, release gates, and branching strategies, and have led DevSecOps maturity assessments that turned ad-hoc deployment processes into governed, auditable release pipelines.

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## PROFESSIONAL EXPERIENCE

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### SRE / DevOps Onboarding Lead — NESO (National Energy System Operator)

Capgemini / Experis · March 2025 – January 2026

- Led SRE and DevOps onboarding for market participants on a highly secure multi-cloud CNI platform: **OpenShift** and **Kubernetes** at the core, hosted across **Azure** (lower environments) and on-premises CNI data centres.
- Operated **F5**, Zscaler, and Checkpoint daily as the CNI edge-protection layer, maintaining WAF policy sets equivalent to **F5 AppProtect** and integrating **OPSWAT MetaDefender** AV/AM scanning into the file-ingress pipeline between Azure test and on-prem production.
- Built and maintained CI/CD delivery through **Azure DevOps Pipelines**, Tekton, and Argo-CD for GitOps-based continuous delivery, bridging the gap between cloud testing environments and on-premises production.
- Ran unified observability with Grafana, Prometheus (+Thanos), Loki, and Jaeger/Kiali across the platform.
- Architected a localised RAG framework (n8n → Qdrant → Claude 3.5 Sonnet) over Bitbucket documentation and Loki logs to speed up root-cause analysis, integrated with JIRA and eChips change-management workflows.
- **Impact:** enhanced grid-balancing stability and streamlined onboarding for new market participants onto CNI infrastructure.

### SRE — National Grid

IBM / NordCloud · March 2024 – March 2025

- Delivered OpenShift automation for a CNI transformation programme, with **RHEL** as the core operating system across the estate, Kubernetes for container orchestration, and VMware for virtualisation.
- Used **Ansible playbooks** to automate RHEL server build, patching, and configuration-drift remediation, replacing manual runbooks and materially cutting provisioning time.
- Operated **F5**, Zscaler, and Checkpoint daily on-premises for CNI network security and WAF policy enforcement.

- Built Tekton pipelines and Argo-CD GitOps workflows for OpenShift continuous delivery; used Bitbucket for version control and Artifactory for artefact management.
- Architected a localised Retrieval-Augmented Generation framework (n8n ingesting Loki logs and Bitbucket docs into a Qdrant vector database, served by Anthropic Claude) to automate technical support and root-cause analysis, integrated into existing Argo-CD/JIRA workflows.
- **Impact:** improved operational efficiency and reduced transformation costs for a Critical National Infrastructure programme.

## DevOps Release Train Engineer — Goldman Sachs

**ACLDigital / Alten / Cprime / Omantel** · Oman, Muscat (onsite) · August 2023 – March 2024

- Assessed release practices against a DevSecOps Maturity Matrix and built a new CI/CD framework across OpenShift (Kubernetes), on-premises gateways, and Azure Sentinel.
- Delivered **Azure DevOps**-driven CI/CD across GCP, Azure, and AWS; integrated Cloud Build on GCP to streamline deployments.
- Used Logic Apps and Azure Virtual Desktop/VDI to serve a population of 4M+ users, including Azure subscription provisioning.
- **Impact:** improved release cadence and reduced manual deployment errors across a regulated financial services environment.

## DevOps Consultant — Ypsomed Switzerland

**Hybrid Contract** · April 2022 – April 2023

- Planned and executed a Git/TFS/VSS migration to **Azure DevOps** for a regulated healthcare device manufacturer, producing full documentation to meet healthcare regulatory requirements.
- Worked across Git, TFS, VSS, VMware, and Citrix, using GitHub Copilot to accelerate legacy code refactoring during the migration.
- **Impact:** successful migration to Git and Azure DevOps with full regulatory compliance and zero legacy-system rollback.

## DevOps Engineer — Cushman & Wakefield

**Remote Contract** · May 2022 – January 2023

- Implemented Infrastructure as Code for a data environment with no existing automated provisioning, using ARM templates and Bicep to deploy to **Azure**.
- Built and maintained **Azure DevOps** pipelines for practical, DSL-based deployments, incorporating AI-assisted IaC validation.
- **Impact:** automated infrastructure provisioning, removing manual setup time from the delivery process.

## DevOps Engineer — Emerson

**Remote Contract** · February 2022 – May 2022

- Solved teams getting stuck at docker-compose by building on-premise Kubernetes (mini-kube) training material and live debugging sessions for multiple stakeholders.
- Used **Bash** and **Python** scripting alongside Git and **Azure DevOps** to support on-premise Kubernetes debugging workflows.
- **Impact:** empowered development teams to deploy and debug on Kubernetes independently, removing a delivery bottleneck.

## Senior DevSecOps Engineer — Concentra Analytics

**Onsite Contract** · April 2019 – November 2019

- Worked with orchestrators (Puppet, Chef) and cloud-native services (CloudFormation, S3, EC2, Lambda, API Gateway, RDS/PostgreSQL, RedShift, KMS) to isolate previously unmaintainable pipeline components.
- Used **Python**, Ruby, and PowerShell for configuration management and container CI/CD pipelines (Docker, ECS/EKS/Kubernetes); administered Linux systems at the command line daily.
- Worked hands-on with networking, routing, IPSec tunnelling, firewalls, ingress/egress, IAM, and encryption for secure pipeline isolation.
- **Impact:** improved resource isolation and simplified pipeline management across a multi-cloud estate.

## Senior DevOps Engineer — Red Box Recorders

**Onsite Contract** · October 2018 – December 2018

- Migrated a CQRS microservices platform onto a new on-premise and cloud Kubernetes environment (Vagrant, Docker Compose, VMware, **CentOS**).
- Deployed **NGINX** as the reverse proxy and load balancer in front of the platform, alongside MetalLB and Traefik for ingress orchestration and Terraform/ARM templates for infrastructure provisioning.
- Hardened the CentOS hosts with **SELinux in enforcing mode**, defining custom policy so containerised services ran under mandatory access control rather than disabling protection.
- Drove the migration cutover with **Bash** scripts (start/stop ordering, health-check polling, log capture) and set up ELK (Elasticsearch, Logstash, Kibana) for microservices logging.
- **Impact:** delivered a fully orchestrated CQRS architecture with improved container visibility and troubleshooting.

## DevOps Architect / Site Reliability Engineer — Scottish Enterprise & Craneware

**Onsite Contracts** · August 2017 – June 2018

- Built continuous delivery pipelines in **Azure DevOps (VSTS)** with Docker, Helm, and Kubernetes for multi-environment delivery.
- Implemented real-time monitoring (Application Insights, Log Analytics, Kafka) and ChatOps-driven release visibility.
- **Impact:** established continuous delivery to multiple environments, cutting release friction for both organisations.

## Cloud Solution Architect — Microsoft

**Hybrid** · February 2016 – February 2017

- Ran architectural design sessions and proof-of-concept implementations to overcome cloud-migration resistance across enterprise customers.
  - Delivered on IaaS/PaaS/SaaS patterns using **Azure DevOps (VSTS)**, ARM, PowerShell, Service Fabric, and Terraform.
  - **Impact:** measurable increase in enterprise cloud-resource consumption across engaged accounts.
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## ADDITIONAL CONTRACT ENGAGEMENTS

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Earlier and shorter contract engagements, all delivered through Azure DevOps/VSTS-based CI/CD with Git-based version control:

- **Canada Life – GWLE** (DevSecOps Engineer, May–Jul 2021) — Azure API Management automation, shift-left security.
  - **Theta Trading** (Azure DevSecOps Engineer, Apr–May 2021) — ARM/YAML pipelines for trading systems, Pipeline-as-Code.
  - **Sword** (Azure DevSecOps Engineer, Dec 2020–Apr 2021) — ARM/YAML pipelines for environment consistency.
  - **Pexabo LLC** (DevOps Integration, Apr 2020–Dec 2021) — GCP transcoding pipelines, multi-cloud backup to AWS Glacier.
  - **Ardent Blue Technologies / Lockheed Martin (TransfrVR)** (DevOps Lead/Consultant, 2019–2020) — Perforce/Unity build automation, Oculus VR release pipelines across distributed teams.
  - **MMT Digital** (Senior DevOps Engineer, Jun–Oct 2018) — Docker/Kubernetes migration, Azure, Bash scripting, Terraform.
  - **Smart Space Software** (Senior DevOps Engineer, Dec 2018–Apr 2019) — Service Fabric migration, PowerShell/**Python** scripting for CI.
  - **Accenture** (Manager, 2015–2016) — SDLC and continuous-delivery management for enterprise Java platforms.
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## TECHNICAL SKILLS

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**Web/Proxy & Application Security:** NGINX (reverse proxy, load balancer, SSL termination), F5 (WAF policy operation, AppProtect-class rule sets), OPSWAT MetaDefender (AV/AM scanning integration), Zscaler, Checkpoint, Cloudflare, Traefik

**Linux & Platform Hardening:** RHEL, CentOS, SELinux (enforcing-mode policy), Linux/Unix administration, systemd, networking (TCP/IP, routing, firewalls, IPSec)

**Automation & Scripting:** Ansible, Bash, Python, PowerShell, Terraform, ARM/Bicep templates

**CI/CD & DevOps:** Azure DevOps (VSTS), GitHub Actions, Jenkins, TeamCity, Tekton, Argo-CD, GitOps

**Containers & Orchestration:** Kubernetes, OpenShift, Docker, Helm, Kustomize, MetalLB

**Cloud Platforms:** Azure, AWS, GCP

**Observability:** Grafana, Prometheus, Thanos, Loki, Jaeger, Kiali, ELK Stack

**Identity & Security:** Active Directory, Azure AD, Keycloak, Kerberos, OAuth2/OIDC, Policy as Code

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## CERTIFICATIONS & TRAINING

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📜 **Claude Architect Professional Certification** (Anthropic) 📜 **Microsoft Certified Architect in Cloud Solutions** (70-532) 📜 MCPS: Microsoft Certified Professional — Developing Microsoft Azure Solutions 📜 Mastering Ansible 📜 Learning Terraform 📜 Learning Linux Command Line 📜 Kubernetes with CI/CD Solutions · Kubernetes: Microservices 📜 AWS CloudFormation (Infrastructure as Code) 📜 Programming Using Python 📜 DevOps Foundations: DevSecOps

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## EDUCATION

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🎓 **Bachelor of Science, Computer Information Technology** Southern New Hampshire University, USA  
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🎓 Severna Park High School, Maryland, USA 🇺🇸

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## SECURITY CLEARANCES

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🔒 **NATO** — Active 🔒 **SC (Security Check)** — Active 🔒 **CNI (Critical National Infrastructure)** — Active  
🔒 **BPSS / NSV** — Active ✓ Background-checked via Sterling, IBM, and Capgemini

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*References and detailed project portfolios available on request.*